SG No. 178 D_6^2 $P6_122$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `6a`, site symmetry: `.2.`

Table 1: Wyckoff bond: `6a@6a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$[x,\ 0,\ 0]$	`[1,-8]`
2	$[-2X,\ -X,\ Z]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[2,-7]`
3	$[X,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[3,-9]`
4	$[-X,\ -2X,\ Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,-11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ -x,\ \frac{5}{6}\right]$	`[5,-10]`
6	$[-X,\ X,\ Z]$	$\left[x,\ x,\ \frac{1}{6}\right]$	`[6,-12]`

Table 2: Wyckoff bond: `6b@6a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[x,\ 0,\ 0]$	`[1,8]`
2	$[0,\ X,\ 0]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[2,7]`
3	$[-X,\ -X,\ 0]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[3,9]`
4	$[-X,\ 0,\ 0]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,11]`
5	$[0,\ -X,\ 0]$	$\left[0,\ -x,\ \frac{5}{6}\right]$	`[5,10]`
6	$[X,\ X,\ 0]$	$\left[x,\ x,\ \frac{1}{6}\right]$	`[6,12]`

Table 3: Wyckoff bond: `12c@6a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ 0,\ 0]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ -x,\ \frac{5}{6}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[x,\ x,\ \frac{1}{6}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[x,\ 0,\ 0]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ -x,\ \frac{5}{6}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ x,\ \frac{1}{6}\right]$	`[12]`

* Wyckoff site: `6b`, site symmetry: `..2`

Table 4: Wyckoff bond: `6a@6b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[-2x,\ -x,\ \frac{7}{12}\right]$	`[2,-10]`

continued ...

Table 4

No.	vector	center	mapping
3	$[-X, -X, Z]$	$\left[x, -x, \frac{11}{12}\right]$	[3,-12]
4	$[-X, 0, Z]$	$\left[-x, -2x, \frac{3}{4}\right]$	[4,-8]
5	$[0, -X, Z]$	$\left[2x, x, \frac{1}{12}\right]$	[5,-7]
6	$[X, X, Z]$	$\left[-x, x, \frac{5}{12}\right]$	[6,-9]

Table 5: Wyckoff bond: 6b@6b, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[x, 2x, \frac{1}{4}\right]$	[1,11]
2	$[-2X, -X, 0]$	$\left[-2x, -x, \frac{7}{12}\right]$	[2,10]
3	$[X, -X, 0]$	$\left[x, -x, \frac{11}{12}\right]$	[3,12]
4	$[-X, -2X, 0]$	$\left[-x, -2x, \frac{3}{4}\right]$	[4,8]
5	$[2X, X, 0]$	$\left[2x, x, \frac{1}{12}\right]$	[5,7]
6	$[-X, X, 0]$	$\left[-x, x, \frac{5}{12}\right]$	[6,9]

Table 6: Wyckoff bond: 12c@6b, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 2x, \frac{1}{4}\right]$	[1]
2	$[-Y, X-Y, Z]$	$\left[-2x, -x, \frac{7}{12}\right]$	[2]
3	$[-X+Y, -X, Z]$	$\left[x, -x, \frac{11}{12}\right]$	[3]
4	$[-X, -Y, Z]$	$\left[-x, -2x, \frac{3}{4}\right]$	[4]
5	$[Y, -X+Y, Z]$	$\left[2x, x, \frac{1}{12}\right]$	[5]
6	$[X-Y, X, Z]$	$\left[-x, x, \frac{5}{12}\right]$	[6]
7	$[Y, X, -Z]$	$\left[2x, x, \frac{1}{12}\right]$	[7]
8	$[X-Y, -Y, -Z]$	$\left[-x, -2x, \frac{3}{4}\right]$	[8]
9	$[-X, -X+Y, -Z]$	$\left[-x, x, \frac{5}{12}\right]$	[9]
10	$[-Y, -X, -Z]$	$\left[-2x, -x, \frac{7}{12}\right]$	[10]
11	$[-X+Y, Y, -Z]$	$\left[x, 2x, \frac{1}{4}\right]$	[11]
12	$[X, X-Y, -Z]$	$\left[x, -x, \frac{11}{12}\right]$	[12]

* Wyckoff site: 12c, site symmetry: 1

Table 7: Wyckoff bond: 12a@12c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-Y, X-Y, Z]$	$\left[-y, x-y, z+\frac{1}{3}\right]$	[2]
3	$[-X+Y, -X, Z]$	$\left[-x+y, -x, z+\frac{2}{3}\right]$	[3]
4	$[-X, -Y, Z]$	$\left[-x, -y, z+\frac{1}{2}\right]$	[4]
5	$[Y, -X+Y, Z]$	$\left[y, -x+y, z+\frac{5}{6}\right]$	[5]
6	$[X-Y, X, Z]$	$\left[x-y, x, z+\frac{1}{6}\right]$	[6]
7	$[Y, X, -Z]$	$\left[y, x, \frac{1}{3}-z\right]$	[7]
8	$[X-Y, -Y, -Z]$	$[x-y, -y, -z]$	[8]

continued ...

Table 7

No.	vector	center	mapping
9	$[-X,\, -X+Y,\, -Z]$	$\left[-x,\, -x+y,\, \frac{2}{3}-z\right]$	[9]
10	$[-Y,\, -X,\, -Z]$	$\left[-y,\, -x,\, \frac{5}{6}-z\right]$	[10]
11	$[-X+Y,\, Y,\, -Z]$	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	[11]
12	$[X,\, X-Y,\, -Z]$	$\left[x,\, x-y,\, \frac{1}{6}-z\right]$	[12]